

Effect of Rear-Wing Angle on F1 Car Performance

Project Scope & Methodology

- An analytical simulation of a representative Formula 1-style vehicle.
- Rear-wing sweep: 0–30° on three synthetic circuit archetypes.
- Corrected force-feasible lap solver and full-bound active-aero search.
- Verified numerical results; not CFD, measured telemetry, or a team setup.

Project Code	ME3106 Motorsport Sim
Model Version	1.0.0
Run ID	9b1c3a7968173ace
Angle Sweep	0.0° to 30.0° (1° step)
Circuits	Low-DF, Balanced, High-DF
Classification	Representative Educational

Research Question and Objectives

- How does rear-wing incidence affect aerodynamic forces and minimum lap time?
- Quantify the trade-off between straight-line resistance and cornering grip.
- Compare low-downforce, balanced, and high-downforce circuit geometries.
- Compare a fixed wing with ideal and actuator-limited two-state strategies.

Key Metrics & Specifications

Independent variable	Rear-wing angle
Main objective	Minimum feasible lap time
Secondary outputs	Speed and tractive energy
Comparison	Same car; different tracks

Study Scope and Modeling Assumptions

- Quasi-steady point-mass motion with front/rear axle load resolution.
- Analytical rear-wing polar combined with a fixed baseline full-car map.
- Load-sensitive tyres, combined grip, longitudinal load transfer, torque and gearing.
- No tyre-temperature dynamics, suspension transients, driver telemetry, or CFD.

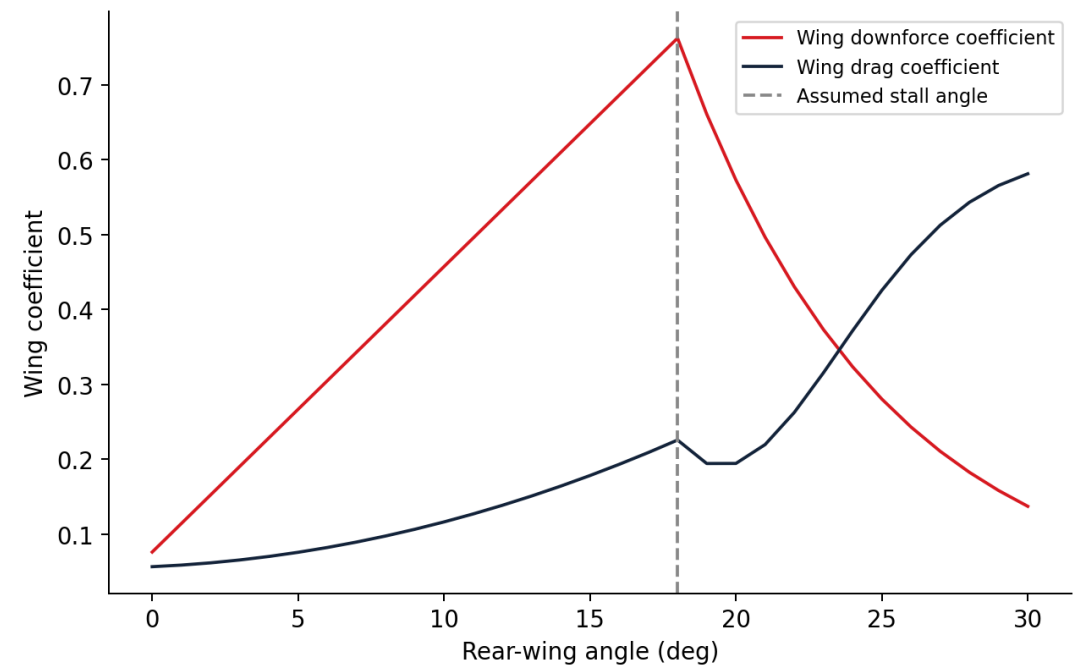
Key Metrics & Specifications

Model scale	Full-car forces
Track spacing	No more than 5 m
Vehicle mass	798 kg
Wheelbase	3.60 m

Aerodynamic Model: Lift, Drag and Stall

- Finite-wing slope: $a = a_0 / [1 + a_0 / (\pi e AR)]$.
- Wing drag includes profile, induced and assumed post-stall terms.
- Pre-stall lift rises linearly; post-stall lift decays exponentially.
- The join is continuous, but its slope can kink at the assumed stall angle.

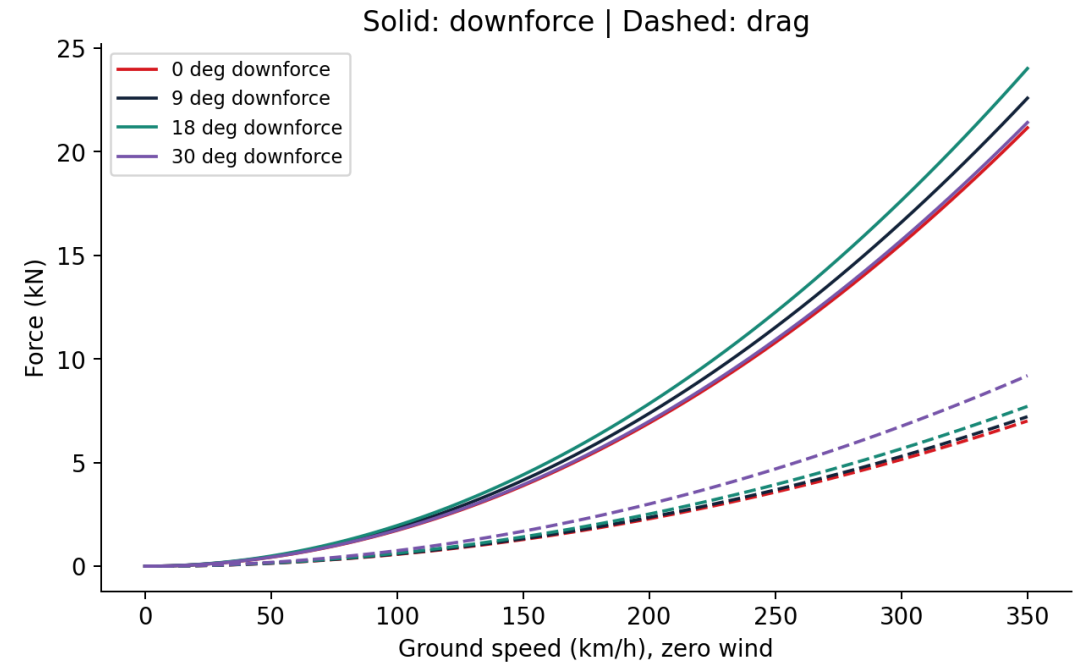
Angle bounds	0–30°
Assumed stall	18°
Aspect ratio	Span squared / wing area



Corrected model output; representative educational vehicle.

Speed-Dependent Downforce and Drag

- Dynamic pressure: $q = 0.5 \rho V_{\text{air}}^2$.
- Downforce = $q A C_{L,\text{total}}$; drag = $q A C_{D,\text{total}}$.
- Relative air speed matters when wind is present.
- Added rear-wing force changes axle loading and the available tyre grip.



Corrected model output; representative educational vehicle.

Air density	1.225 kg/m3 nominal
Reference area	1.50 m2
Comparison angles	0°, 9°, 18°, 30°

Tyre Grip and Longitudinal Load Transfer

- Load sensitivity: $\mu = \mu_{ref} (F_z / F_{z_ref})^{-n}$.
- Combined demand: $(F_x/F_{x,max})^p + (F_y/F_{y,max})^p \leq 1$.
- Lateral demand is normalized by lateral friction, not longitudinal friction.
- Actual acceleration shifts axle normal loads and alters traction/braking capacity.

Key Metrics & Specifications

Tyre exponent n	0.08
Ellipse exponent p	2
Nominal μ_x / μ_y	1.85 / 1.95
Load resolution	Front and rear axles

Powertrain and Force Balance

- Available wheel thrust follows the interpolated engine torque curve and eight gears.
- Resistance combines aerodynamic drag, rolling resistance and track gradient.
- Each segment must satisfy thrust - brake - resistance = mass x acceleration.
- Constant-speed cornering must reserve enough longitudinal grip to overcome drag.

Key Metrics & Specifications

Gear count	8
Driveline efficiency	0.94
Redline	15,000 rpm
Energy metric	Positive wheel tractive work

Closed-Track Lap Simulation Workflow

- Load validated car configuration and discretize the track at ≤ 5 m.
- Construct a sustainable local speed envelope for the commanded wing profile.
- Apply forward acceleration and backward braking passes until convergence.
- Include the closing segment, then check achieved acceleration and force closure.

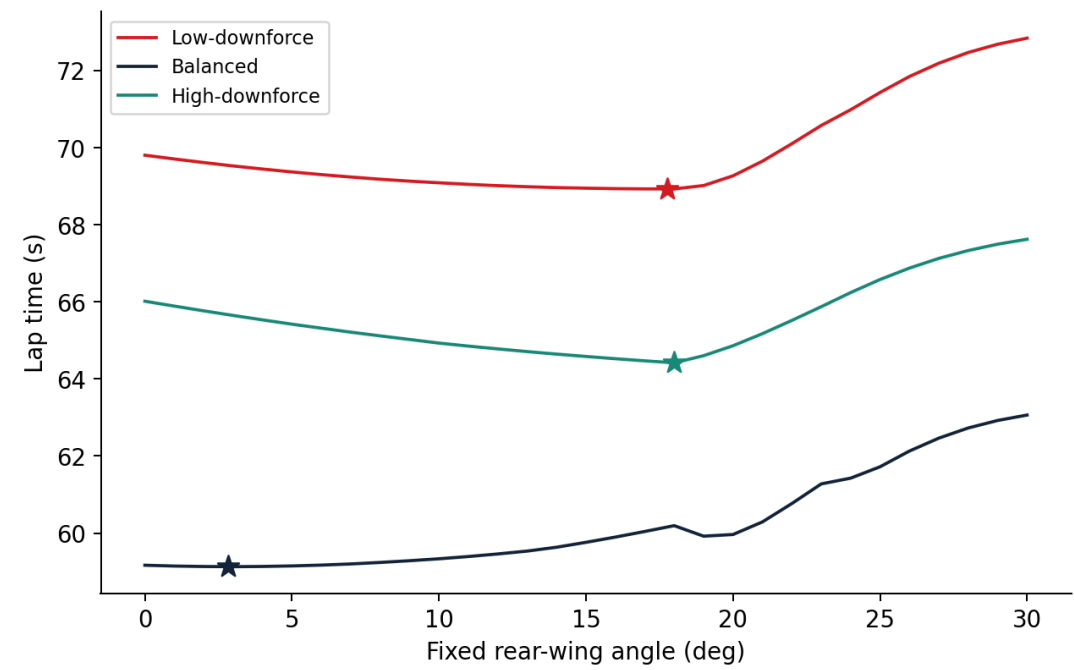
Key Metrics & Specifications

Speed convergence	≤ 0.001 m/s
Iteration ceiling	200
Force feasibility guard	≥ -5 N
Force closure guard	≤ 0.000001 N

Fixed-Wing Sweep Across Three Circuits

- 93 physical laps: 31 integer angles x three synthetic circuits.
- Every case checked for convergence, force feasibility and closed-segment integration.
- All angle cases retain the original 5 m track discretization.
- The best wing setting depends on this car model and the circuit geometry.

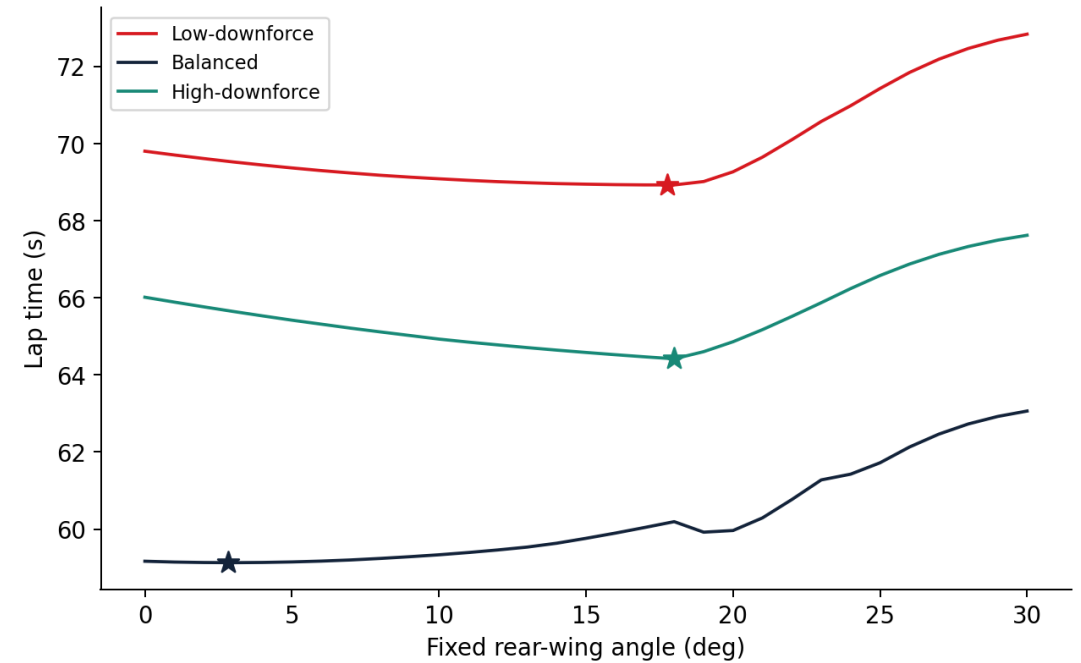
Grid cases	93 / 93 checked
Wing sweep	0°, 1°, ... 30°
Track types	Low-DF / Balanced / High-DF



Corrected model output; representative educational vehicle.

Continuously Refined Fixed-Wing Optima

- Physical lap solves refine candidate basins around the integer sweep.
- Bounds and neighboring integer points remain part of the search.
- Balanced-track optimum is near 2.819° ; the other two are near 18° .
- These are nominal model optima, not uncertainty confidence intervals.

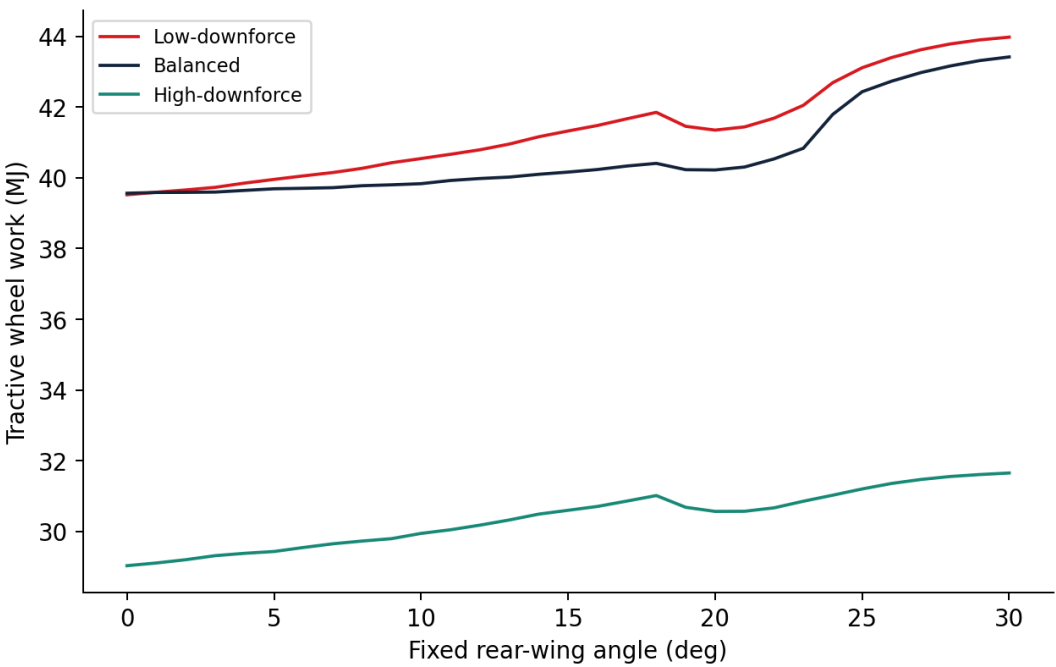


Corrected model output; representative educational vehicle.

Low-DF fixed	17.752° / 68.927 s
Balanced fixed	2.819° / 59.130 s
High-DF fixed	18.000° / 64.418 s

Speed and Energy: Secondary Performance Metrics

- Record maximum speed and positive tractive wheel work for every fixed lap.
- Energy is integrated from achieved tractive force over distance.
- Minimum lap time and minimum tractive energy are different objectives.
- Wheel tractive work is not fuel consumption or battery-energy demand.



Corrected model output; representative educational vehicle.

Source	Fresh 93-case fixed audit
Energy unit	MJ per modeled lap
Speed unit	km/h

Two-State Active Aerodynamics

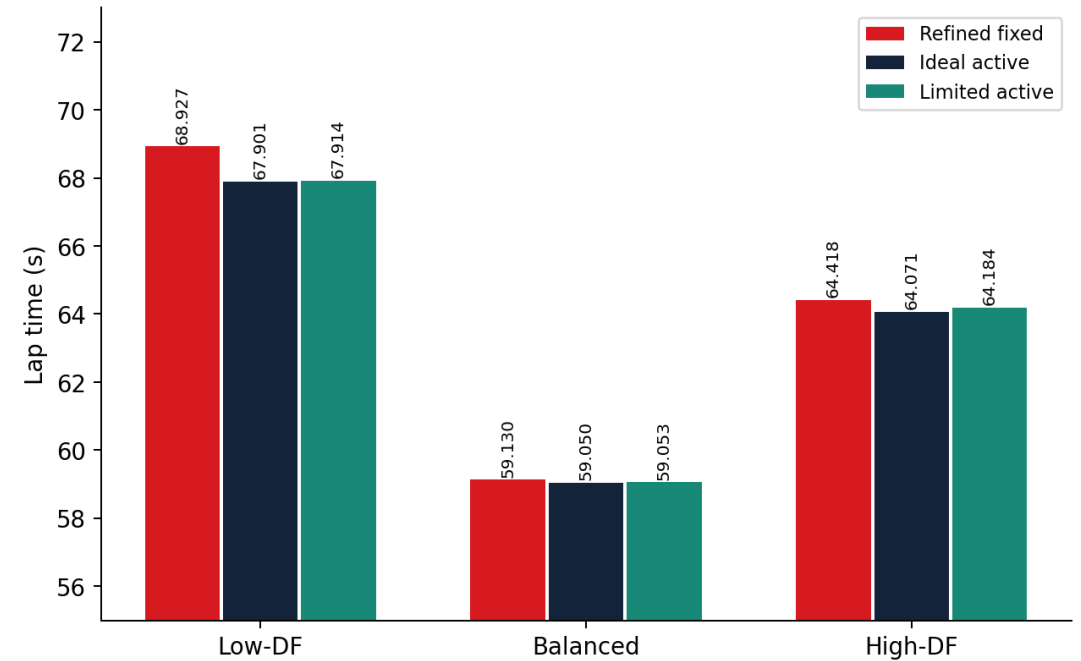
- Use an open, lower-angle state on eligible straight sections.
- Use a closed, higher-angle state under the modeled switching conditions.
- Search every unique integer pair within 0–30°, with $\geq 1^\circ$ separation.
- Compare ideal state changes against finite-rate, delayed actuator behavior.

Key Metrics & Specifications

Pairs per search	465
Actuator limit	90°/s
Activation delay	0.2 s
Deactivation delay	0.1 s

Fixed Versus Ideal and Limited Active Aero

- Compare optimized strategies using the same car and original track meshes.
- Active search evaluates actual physical laps, not a coarsened surrogate.
- All completed cases must satisfy force, convergence and actuator checks.
- Interpret small differences relative to model assumptions and numerical resolution.



Low-DF limited	67.914 s
Balanced limited	59.053 s
High-DF limited	64.184 s

Corrected model output; representative educational vehicle.

Validation and Reproducibility Evidence

- Fixed sweep: 93 checked laps, including force, time and energy closure.
- Active searches: 465 unique pairs per circuit and actuator mode.
- Scalar force checks include the track closing segment and achieved acceleration.
- Source hashes, input hashes, runtime versions and completion receipts preserve identity.

Key Metrics & Specifications

Active scope	3 circuits x 2 modes
Active cases	2,790 total
Audit type	Numerical feasibility
Empirical validation	Not performed

MATLAB and Simulink Implementation

- MATLAB source includes configuration, aero, dynamics and fixed-wing lap analysis.
- Simulink model/builder is supplied for a speed-dependent aero/actuator demonstration.
- The aero block chain is: speed and angle -> coefficients -> forces -> scopes.
- Full native active/uncertainty/sensitivity execution and strict runtime parity remain unverified.

Key Metrics & Specifications

MATLAB entry point	matlab/run_project.m
Parity tolerance	1e-4 required
Simulink scope	Aero/actuator demonstration
Runtime status	Full native validation pending

Uncertainty and Sensitivity: What Remains

- A validated $\geq 1,000$ -sample, nine-parameter uncertainty study is still required.
- The tested uncertainty surrogate failed independent validation and was rejected.
- Direct full-physics Monte Carlo is selected, but not yet implemented or executed.
- No confidence intervals or sensitivity rankings are claimed in this presentation.

Key Metrics & Specifications

Planned random seed	3106
Planned samples	$\geq 1,000$ per circuit
Accepted uncertainty	None yet
Legacy MC figures	Excluded

Main Findings and Engineering Interpretation

- There is no universal wing angle: optimize for the complete circuit, not one aerodynamic metric.
- Correct force balance changes the accepted lap speeds, optima and strategy comparisons.
- Two-state active aero provides a nominal model benefit when feasible switching is allowed.
- Final empirical calibration, native parity and validated uncertainty are necessary next steps.

Key Metrics & Specifications

Nominal fixed optimum	Circuit dependent
Control takeaway	Respect actuator limits
Model takeaway	Reserve grip for resistance
Claim boundary	Educational simulation

Evidence, Source Files and Questions

- Inputs: config/project.json and tracks/*.json define assumptions and geometry.
- Physics: src/f1wing/aero.py, dynamics.py, lap.py and strategies.py.
- Corrected results: docs/verification/*-20261005.csv and matching JSON audit receipts.
- Native files: matlab/ and results/simulink/; incomplete runtime scope is disclosed.

Key Metrics & Specifications

Presentation	Editable PowerPoint + PDF
Speaker support	Notes and slide-by-slide guide
Numerical source	Corrected audit receipts
Questions	Model, methods, trade-offs